

Time-series Foundation Model Evaluation

Executive Summary

Thesis project

Evidence checked through 27 August 2026

Decision snapshot

No synchronized evaluation is eligible evidence for the current CSV-only, replacement-exclusion contract. The completed manifests consumed prepared tensor files and, for controls, retained the former additive exclusions. Their measurements are historical signals only; the affected identities must be rerun before supporting a current scientific decision.

1 Evidence inventory

Result object	Count
Completed pre-contract manifests	76
Univariate / controls / covariates / foundation	60 / 9 / 3 / 4
Manifests still marked running	2
Interrupted foundation manifests	1
Current-contract completed manifests	0

2 Obtained evidence

Historically, the single-seed full-univariate snapshot evaluated Electricity, Solar, and Traffic at 168:24, 336:48, and 504:168 with stride 512. Chronos-2 beat the best non-trainable baseline in nMSE on all nine settings; the mean improvements were 31.335%, 43.099%, and 37.394%, respectively. These identities consumed tensor inputs and therefore cannot be reused now. The former Exchange Rate settings also used hourly scales on daily data.

At Electricity 504:168, the historical four-model test did not establish a metric-independent winner: Chronos-Bolt had the lowest nMSE, while TS-ICL had the lowest raw MSE. The covariate family showed a large Chronos-2 identity signal, but only three configurations completed; controls stopped after nine. Both families are incomplete and pre-contract, so neither supports a current conclusion. Two obsolete manifests remain marked running under Selena jobs 2910738 and 2910739, and DGX job 45673 was interrupted by missing Chronos-2 weights.

3 Next evidence

Publish and synchronize the current CSV-only code and dataset configuration. Discard the obsolete running manifests and rerun each test family so its manifest records the current input contract and effective exclusions. After those boundaries pass, run cadence-aware full univariate, controls, covariates, and foundation profiles with Chronos-2, Chronos-Bolt, Chronos-T5, and TS-ICL, including daily Exchange Rate and feasible TIME datasets, before making new claims.